

Jiro David

Software Engineer

London, UK | 07555 979116 | jiro david153@gmail.com | linkedin.com/in/jirodavid | github.com/JiroDavid | jirodavid.dev

Education

University of the Arts London

BSc (Hons), Computer Science

Sep 2023 – Jul 2026

London, UK

- First Class Honours
- Exhibited TwitchTok at UAL CCI Degree Show 2026, presenting to industry professionals, recruiters, and the public

Projects

TwitchTok

- Python, FastAPI, Next.js, React, TypeScript, Tailwind CSS, Whisper, Ollama, LLaVA, Llama 3, FFmpeg, Twitch OAuth
- Full-stack web application converting Twitch VODs to vertical 9:16 short-form clips; adopted by active streamers following a beta.
- Formally evaluated against a manual baseline: all participants completed the task with AI enabled; none finished within the time limit in manual mode.
- Built subtitle generation via OpenAI Whisper, automatic layout detection via LLaVA through Ollama, three configurable layout modes, a live caption editor, and automated metadata generation via Twitch OAuth.
- Implemented FastAPI and Uvicorn backend with Next.js 16, React 19, and TypeScript frontend; FFmpeg and yt-dlp for video processing.

ShimArchives

- Python, FastAPI, RAG, Vector Embeddings, Twitch API, LLM, PostgreSQL
- RAG system ingesting Twitch API data across VOD transcripts, chat logs, and clip metadata into a structured vector store.
- Conversational interface enabling natural language queries such as “when did I say this on stream”, returning accurate timestamps and source references.
- Designed ingestion pipeline with embedding strategy across heterogeneous data types; implemented grounding logic to prevent hallucinated outputs.
- Diagnosed and resolved a complex chat replay sync issue: Twitch comment offsets anchor to live stream time, not edited VOD time, requiring re-anchoring against HLS discontinuity markers.

ShimaMMR

- Python, discord.py, SQLAlchemy (async), Alembic, SQLite, httpx, pytest-asyncio, Docker
- Production Discord bot providing Elo-based MMR, rank tiers, and leaderboards for a 100+ member Valorant community’s nightly customs and scrims.
- Built a performance-adjusted rating engine (dynamic K-factor, combat-score modifier, loss-streak dampening) tuned for 10-player pickup matches, not generic 1v1 Elo.
- Integrated the HenrikDev Valorant API for automatic match detection, including roster-heuristic match classification, rate-limit back-off, and human-in-the-loop resolution for unrecognized players.
- Implemented a confirm/dispute workflow with admin correction and recompute cascades to keep MMR data consistent; containerized with Docker Compose, 140+ tests.

ShimaTTS

- Python, FastAPI, F5-TTS, PyTorch, pywebview, Twitch EventSub API
- Local Twitch TTS desktop application with AI voice cloning; fully on-device, no cloud dependency.
- Clones a voice from a 10-30 second audio sample using F5-TTS on GPU (PyTorch); channel point redemptions spoken in the cloned voice in real time.
- Implemented Twitch EventSub WebSocket integration for real-time event handling with animated OBS browser-source overlay.
- Built native desktop UI with pywebview and FastAPI backend; includes voice and GIF libraries, TTS queue for async processing, and system tray support.

Video Storyboarder

- Python, Gradio, Whisper, Ollama, LLaVA, Llama 3, ComfyUI, Stable Diffusion XL, LoRA
- Pipeline generating a structured storyboard and teleprompter script from a YouTube creator’s video intro.
- Transcribes audio with Whisper, analyses video frames with LLaVA via Ollama, and generates storyboard content with Llama 3.
- Renders sketch-style storyboard panels using ComfyUI with SDXL Base and a custom-trained LoRA fine-tuned on storyboard sketch imagery.
- Served via a Gradio web UI with structured JSON and Markdown output.

Bindings of NopeYep

- JavaScript, HTML5 Canvas, Web Audio API
- Browser game built in vanilla JS and HTML5 Canvas; shipped as a single self-contained HTML file with no build tooling or dependencies.
- Delta-time game loop with DPR-aware Canvas renderer, particle system, and procedural audio via Web Audio API.
- Multiple enemy types with distinct AI state machines, a phased boss fight with telegraph mechanics, and room-based progression with skill upgrades.

Experience

Creative Director & Social Media Manager

2023 – 2026

Freelance

London, UK

- Managed content pipelines and production teams for multiple large-scale Twitch streamers simultaneously across North American time zones.
- Edited short and long form video content across TikTok, YouTube Shorts, Instagram Reels, and full YouTube videos end to end, from raw footage to final upload.
- Assigned and distributed editing projects across teams, hired editors for specific projects, and reviewed output before delivery to streamers.
- Produced on-screen stream overlays and interactive widgets using Pogly, contributing to live production quality and viewer experience.
- Developed stream concepts and creative ideas directly with streamers, contributing to content strategy and on-screen presentation.
- Directly informed development of TwitchTok and ShimArchives by identifying bottlenecks in existing workflows and building automated solutions.

Granton Primary School

2023

Teaching & Administrative Assistant

London, UK

- Supported classroom activities and assisted teaching staff in a fast-paced educational environment.
- Communicated with children, parents, and colleagues across a range of situations, adjusting tone and approach as needed.
- Managed multiple administrative workflows and maintained organised records with consistent attention to detail.

Lutong Pinoy Filipino Restaurant

2024

Kitchen & Front of House Assistant

London, UK

- Assisted with food preparation, serving customers, and general kitchen duties in a busy restaurant environment.
- Supported the team during service, maintaining a clean workspace and following food hygiene standards.

Technical Skills

- **Languages:** Python, TypeScript, JavaScript, HTML, CSS, SQL, Java, Node.js
- **Frameworks & Libraries:** FastAPI, Next.js, React, Tailwind CSS, Gradio, Pydantic, discord.py, SQLAlchemy, httpx
- **AI & ML:** OpenAI Whisper, F5-TTS, Ollama, LLaVA, Llama 3, ComfyUI, Stable Diffusion XL, LoRA fine-tuning, RAG, Vector Embeddings
- **Tools & Platforms:** Git, GitHub, FFmpeg, yt-dlp, REST APIs, OAuth 2.0, Docker, Alembic, SQLite, pytest, Claude Code
- **Specialisation:** LLM Pipelines, Multi-Model Orchestration, Computer Vision, API Design, Full-Stack Development, Agentic AI